NAME: _____ Geometry (FL B.E.S.T.)

11.2 Bisecting Segments and Angles

To **bisect** means to split exactly in half. This worksheet builds the perpendicular bisector of a segment and the bisector of an angle using compass and straightedge, then proves — not just states — the two equidistance theorems that make these constructions useful: every point on a segment's perpendicular bisector is equidistant from the segment's endpoints, and every point on an angle's bisector is equidistant from the angle's two sides. These constructions and theorems are the direct foundation for locating a triangle's circumcenter and incenter in the next lesson.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. For the perpendicular bisector of $\overline{AB}$, the first arcs are centered where?

2. $AB = 20$. Find AM after the perpendicular bisector is constructed.

3. After the vertex arc crosses both rays, where are the next equal-radius arcs centered?

4. $m\angle ABC = 70°$ is bisected. Find each half.

5. A point on the perpendicular bisector of $\overline{CD}$ is equidistant from which two points?

6. A point on an angle's bisector is equidistant from what?

7. At what angle does a perpendicular bisector meet the original segment?

8. An angle $(6x)°$ is bisected into two angles of $(2x + 6)°$ each. Find x.

9. Must the compass radius for a perpendicular bisector exceed $\frac{1}{2}AB$? Why?

10. What two things does bisecting a segment create?

11. What does bisecting an angle create?

12. How many intersection points of the equal-radius arcs are needed to draw the perpendicular bisector line?

13. Define "perpendicular bisector."

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14. M is the midpoint of $\overline{AB}$, with $AM = 2x + 3$ and $AB = 5x - 2$. Find x and AB.

15. In $\triangle ABC$, the perpendicular bisector of $\overline{BC}$ turns out to be the exact same line as the bisector of $\angle A$. What must be true about $\triangle ABC$, and why?

✓ Answer Key

1. At A and at B, using the same radius
2. 10
3. At the two points where the vertex arc met the rays
4. 35°
5. C and D
6. The two sides (rays) of the angle
7. 90°
8. $x = 6$
9. Yes — otherwise the two arcs cannot intersect
10. Two congruent segments and a midpoint
11. Two congruent angles
12. Two (one on each side of $\overline{AB}$)
13. A line through a segment's midpoint, perpendicular to the segment
14. $x = 8$; $AB = 38$
15. $\triangle ABC$ is isosceles with $AB = AC$ — by symmetry, the angle bisector from the apex of an isosceles triangle also perpendicularly bisects the base